3DIC: Cu-Cu Direct Bonding

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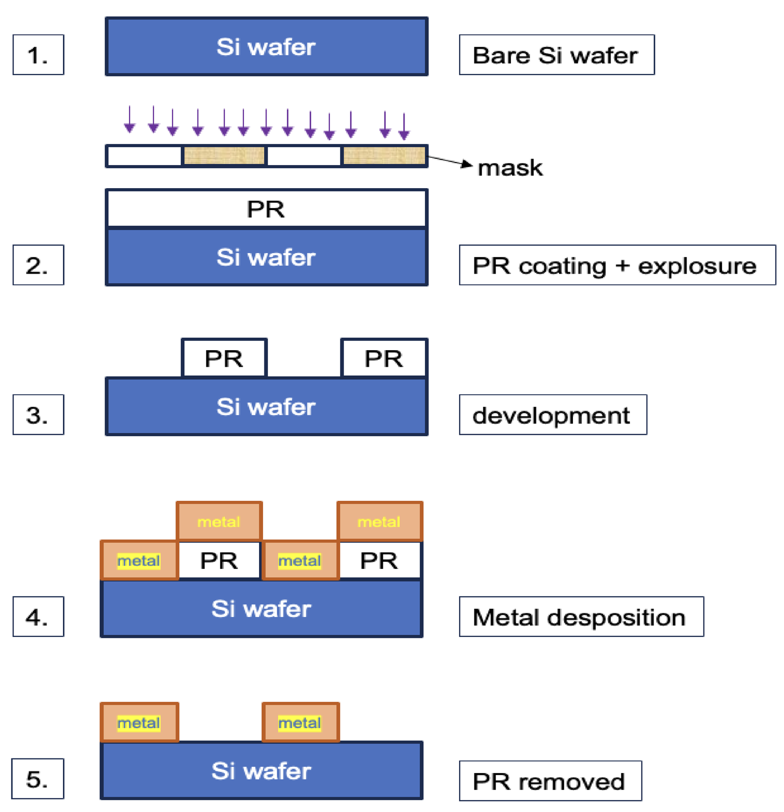
Abstract

This research presents study on the manufacturing process of Cu-Cu direct bonding for 3DIC. Main procedures include RCA cleaning, SiO2, film growth, lithography, E-gun deposition of adhesion layer and Cu film, cleaning and Cu oxide layer removal, dicing, and bonding. Of the two cleaning methods-plasma treatment and acid treatment employed on wafers, the one with plasma treatment demonstrate superior bonding strength.

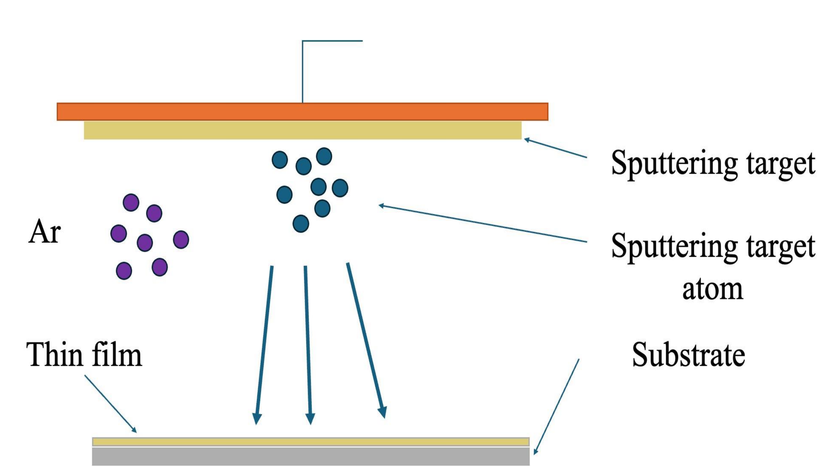
The study results encompass SEM, pull tests, SAT, and TEM examination results on bonding condition of wafer with both plasma and acid treatment. This research offers insights into optimizing the fabrication process for better conductivity and bonding quality.

Keywords: 3DIC, enhanced performance in Cu-Cu direct bonding for 3DIC applications, plasma treatment, acid treatment.

Experimental

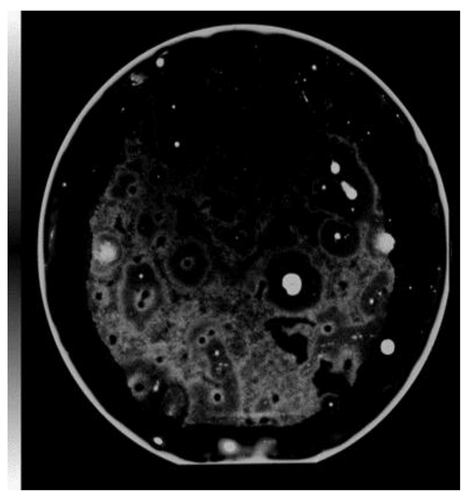


lithography

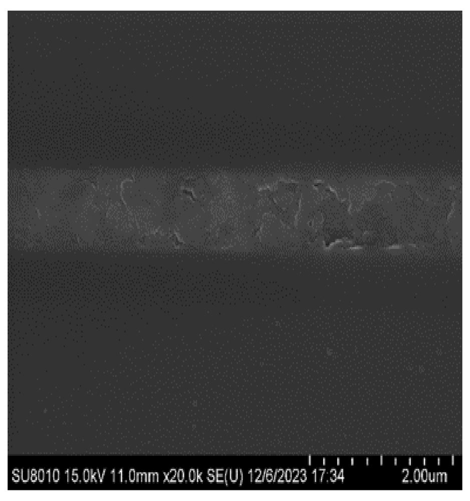


Adhesion layer and copper layer deposition-PVD

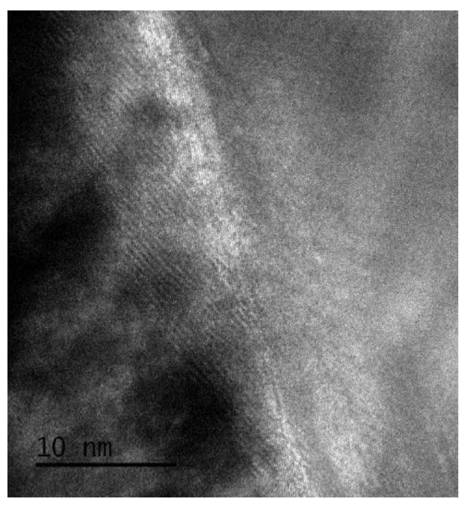
Characterization



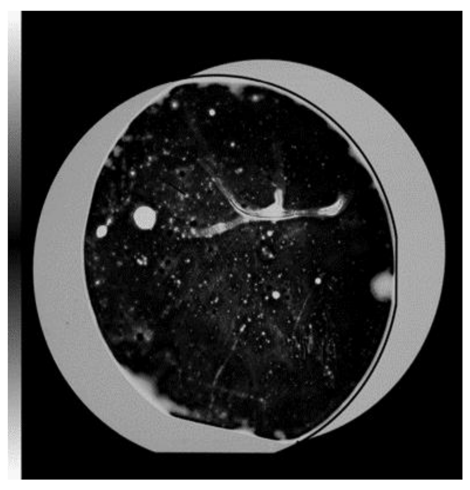
SAT result (plasma)



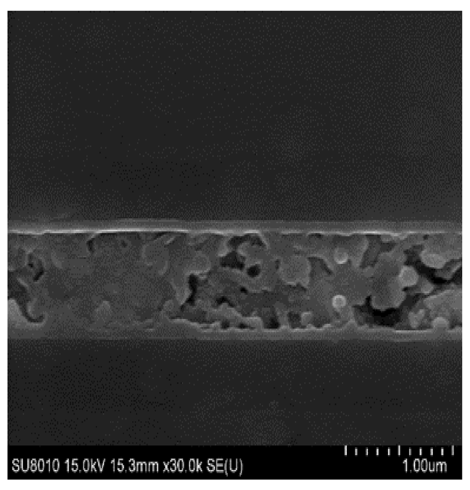
SEM result (plasma)



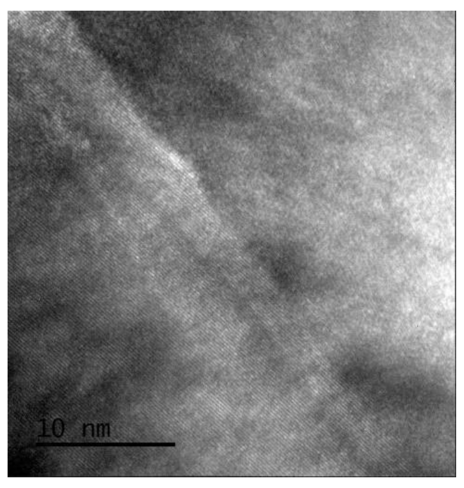
TEM result (plasma)



SAT result (acid)

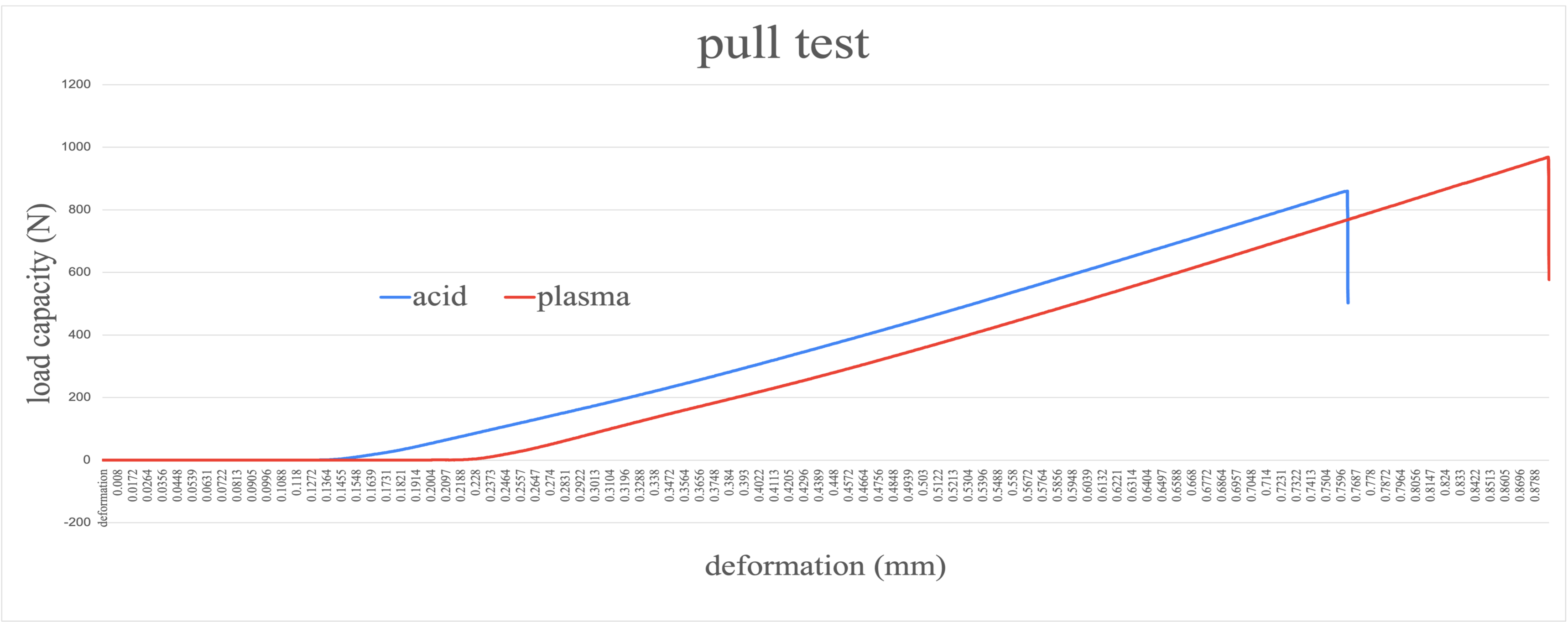


SEM result (acid)

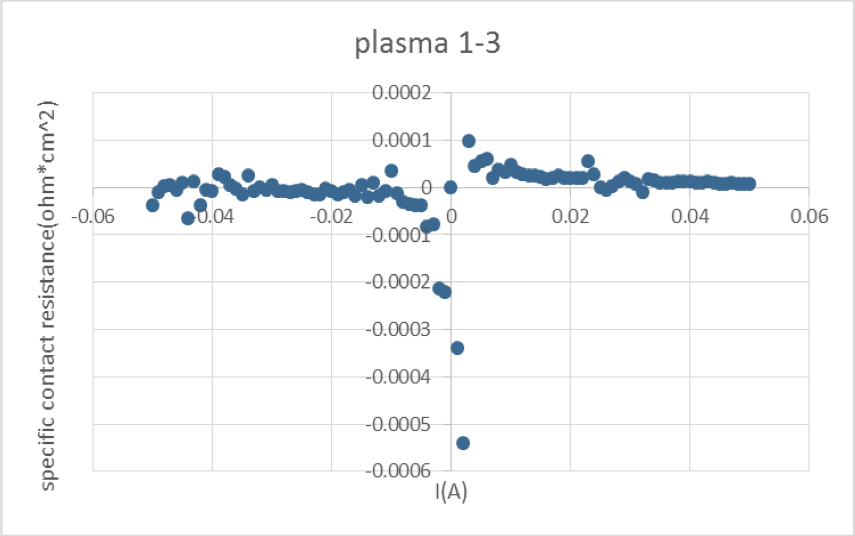


TEM result (acid)

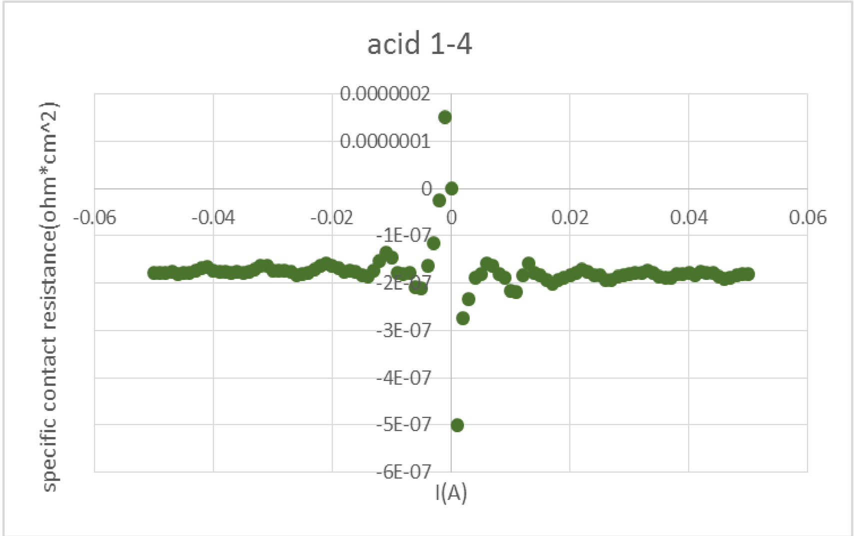
Results



pull test



electrical test (plasma)



electrical test (acid)

Discussion

Based on the testing results, plasma-treated wafers demonstrated stronger bonding quality, as indicated by SEM observations and Pull Test results. However, TEM analysis showed that acid-treated samples had fewer bonding defects, with less cracking and fewer voids. In terms of electrical performance, acid-treated wafers exhibited lower resistance, while plasma-treated wafers experienced circuit damage, making a direct comparison of electrical properties difficult.

In general, plasma treatment provided better bonding strength, while acid treatment resulted in fewer defects and lower resistance. Thus, the choice of cleaning method depends on the specific application requirements of the wafers.